

PRACTICAL ASSIGNMENT 6 REPORT

COLLEGE COURSE & STUDENT MANAGEMENT SYSTEM USING REACT

1. OBJECTIVE / AIM

Develop a modular React application featuring reusable components, client-side routing using React Router, state hooks, attendance warning badge conditional rendering, and academic report compilation.

2. IMPLEMENTATION PROCEDURE

- **Step 1:** Initialized React application with modern Vite bundling and SCSS modular design tokens.
- **Step 2:** Developed interactive components (Navbar, StudentCard, CourseCard, DashboardCard).
- **Step 3:** Established multi-route routing structure (/dashboard, /students, /courses).
- **Step 4:** Automated UI verification and generated formal submission artifacts using ReportCraft MCP.

3. SOURCE CODE LISTINGS

File: src/App.jsx

```
import React, { useState } from 'react';
import Navbar from '../components/Navbar';
import DashboardCard from '../components/DashboardCard';

export default function App() {
  const [activeTab, setActiveTab] = useState('dashboard');
  return (
    <div className="app-container">
      <Navbar active={activeTab} onSelect={setActiveTab} />
      <main className="main-content">
        <h1>College Student & Course Management System</h1>
        <DashboardCard title="Active Enrolments" count={250} />
      </main>
    </div>
  );
}
```

4. OUTPUT SCREENSHOTS

Figure 1: Dashboard Overview Screen (/dashboard)

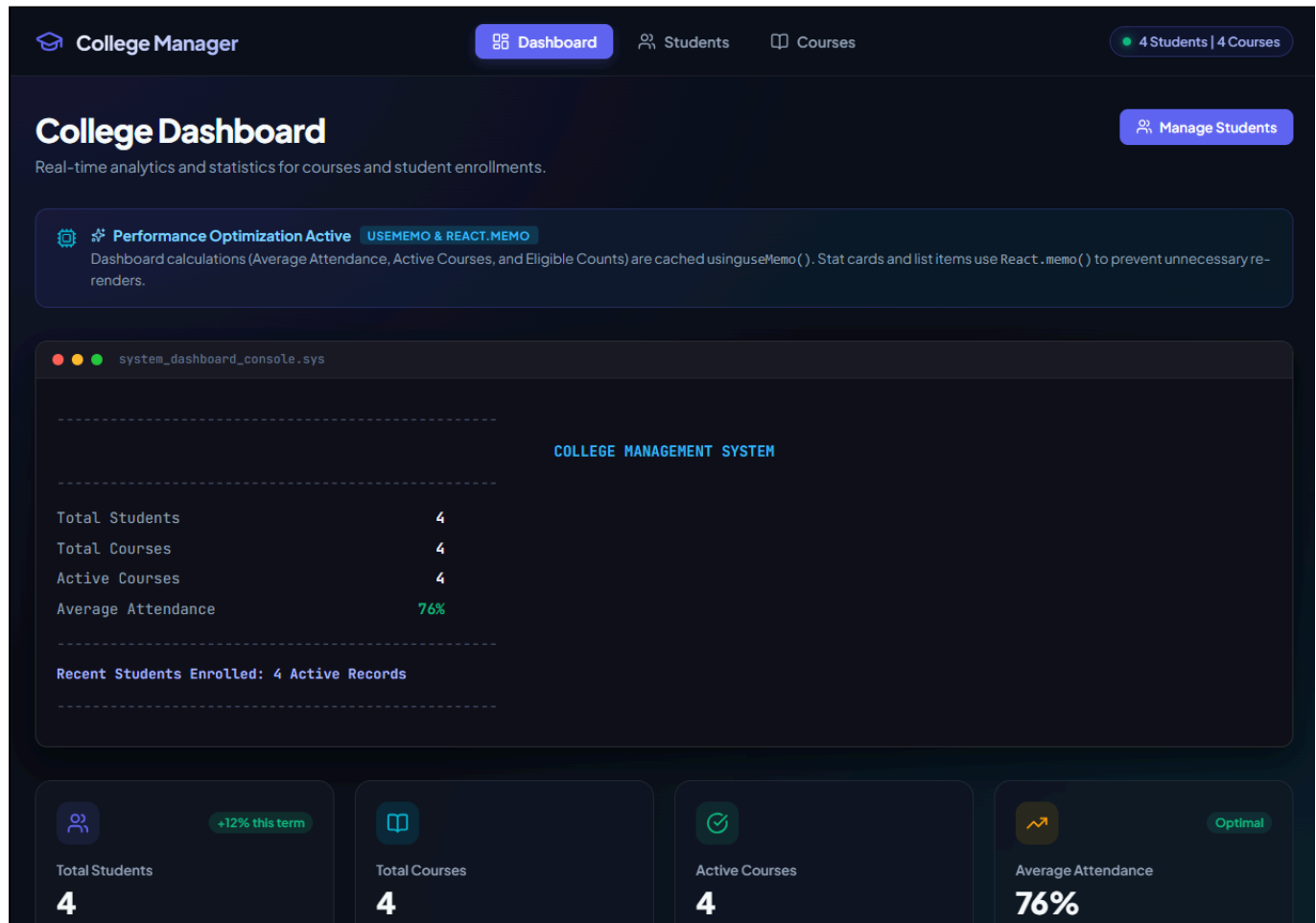


Figure 2: Student Management Directory View (/students)

